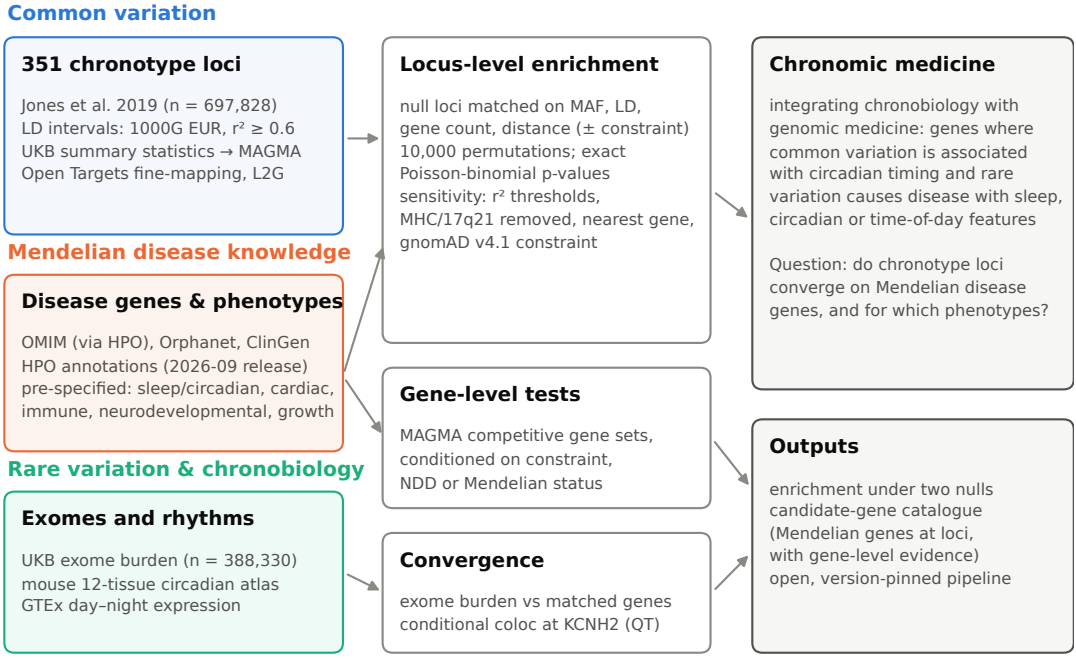
Common variation
351 chronotype loci
Jones et al. 2019 (n = 697,828)
LD intervals: 1000G EUR, r² ≥ 0.6
UKB summary statistics → MAGMA
Open Targets fine-mapping, L2G
Mendelian disease knowledge
Disease genes & phenotypes
OMIM (via HPO), Orphanet, ClinGen
HPO annotations (2026-09 release)
pre-specified: sleep/circadian, cardiac,
immune, neurodevelopmental, growth
Rare variation & chronobiology
Exomes and rhythms
UKB exome burden (n = 388,330)
mouse 12-tissue circadian atlas
GTEx day–night expression
Locus-level enrichment
null loci matched on MAF, LD,
gene count, distance (± constraint)
10,000 permutations; exact
Poisson-binomial p-values
sensitivity: r² thresholds,
MHC/17q21 removed, nearest gene,
gnomAD v4.1 constraint
Gene-level tests
MAGMA competitive gene sets,
conditioned on constraint,
NDD or Mendelian status
Convergence
exome burden vs matched genes
conditional coloc at KCNH2 (QT)
Chronomic medicine
integrating chronobiology with
genomic medicine: genes where
common variation is associated
with circadian timing and rare
variation causes disease with sleep,
circadian or time-of-day features
Question: do chronotype loci
converge on Mendelian disease
genes, and for which phenotypes?
Outputs
enrichment under two nulls
candidate-gene catalogue
(Mendelian genes at loci,
with gene-level evidence)
open, version-pinned pipeline